In [10]:
```
# CELL 0
# OBTAINING AND LOADING DATA
import pandas as pd

# Load the datasets
marketing_ads_data = pd.read_csv("marketing_ads_data.csv")
sales_transaction_data = pd.read_csv("sales_transaction_data.csv")

# Preview
print("=== Marketing Ads Data ===")
print(f"shape: {marketing_ads_data.shape}")
print(f"columns: {marketing_ads_data.columns.tolist()}")
print("\n=== Preview 10 baris pertama ===")
print(marketing_ads_data.head(10).to_string(index=False))

print("\n=== Sales Transaction Data ===")
print(f"shape: {sales_transaction_data.shape}")
print(f"columns: {sales_transaction_data.columns.tolist()}")
print("\n=== Preview 10 baris pertama ===")
print(sales_transaction_data.head(10).to_string(index=False))
```

```
=== Marketing Ads Data ===
shape: (1030, 12)
columns: ['ad_id', 'campaign_name', 'audience_type', 'product_type', 'product_name', 'variant', 'date', 'impressions', 'click
s', 'spend_idr', 'sessions', 'add_to_cart']

=== Preview 10 baris pertama ===
  ad_id        campaign_name audience_type product_type                  product_name        variant         date  impressions
clicks  spend_idr  sessions  add_to_cart
AD_0032   Awareness Campaign          cold        Serum   DermaGlow Brightening Serum   Video_Review   2023-01-06         8060
253   458896.0       216           29
AD_0110   Awareness Campaign          cold    Sunscreen DermaGlow UV Shield Sunscreen   Video_Review   01/09/2023        17314
542   108070.0       452           72
AD_0137 Retargeting Campaign          warm    Sunscreen DermaGlow UV Shield Sunscreen Static_Catalog   2023-04-13        16374
180   135435.0       146            8
AD_0089 Retargeting Campaign          warm    Sunscreen DermaGlow UV Shield Sunscreen Static_Catalog Nov 14, 2023        10735
159   244118.0       128            6
AD_0919  Flash Sale Campaign          cold     Cleanser     DermaGlow Gentle Cleanser   Video_Review Apr 15, 2023        12446
268        NaN       217           36
AD_0169  Flash Sale Campaign          cold        Serum   DermaGlow Brightening Serum   Video_Review   2023-03-29        13731
443   269631.0       355           43
AD_0871 Retargeting Campaign          warm        Serum   DermaGlow Brightening Serum   Video_Review Mar 16, 2023         8834
223   236016.0       178           18
AD_0319   Awareness Campaign          cold        Serum   DermaGlow Brightening Serum   Video_Review   2023-12-09        14569
311   287954.0       254           32
AD_0262 Retargeting Campaign          warm        Serum   DermaGlow Brightening Serum Static_Catalog   2023-08-08         8974
133   248087.0       119           12
AD_0536 Retargeting Campaign          warm        Serum   DermaGlow Brightening Serum Static_Catalog   23/12/2023         8981
127   326453.0       103           16

=== Sales Transaction Data ===
shape: (2500, 8)
columns: ['transaction_id', 'ad_id', 'transaction_date', 'product_type', 'product_name', 'variant', 'revenue_idr', 'order_statu
s']

=== Preview 10 baris pertama ===
transaction_id   ad_id transaction_date product_type                  product_name        variant  revenue_idr order_status
    TXN_001448     NaN       2023-03-15     Cleanser       DermaGlow Gentle Cleanser Static_Catalog     403000.0      Success
    TXN_001115 AD_0958       2023-11-30  Moisturizer DermaGlow Hydrating Moisturizer   Video_Review     187000.0      Success
    TXN_001065 AD_0145       2023-12-15     Cleanser       DermaGlow Gentle Cleanser Static_Catalog          0.0    Cancelled
    TXN_002288 AD_0349       2023-11-08     Cleanser       DermaGlow Gentle Cleanser Static_Catalog    -187000.0    Refunding
    TXN_001538 AD_0317       2023-06-22  Moisturizer DermaGlow Hydrating Moisturizer   Video_Review     236000.0      Success
    TXN_000669 AD_0992       2023-11-15        Serum     DermaGlow Brightening Serum   Video_Review     521000.0      Success
    TXN_001584 AD_0815       2023-10-12    Sunscreen   DermaGlow UV Shield Sunscreen   Video_Review     215000.0      Success
    TXN_002405 AD_0698       2023-07-19     Cleanser       DermaGlow Gentle Cleanser Static_Catalog     264000.0      Success
    TXN_000498     NaN       2023-05-26     Cleanser       DermaGlow Gentle Cleanser   Video_Review          0.0    Cancelled
    TXN_002481 AD_0543       2023-09-06  Moisturizer DermaGlow Hydrating Moisturizer Static_Catalog     132000.0      Success
```

In [2]:
```
# CELL 1
# SCRUBBING AND PREPARING DATA

print("===== MARKETING ADS DATA =====")

# Data Structure
print("\nStruktur Data:")
print(marketing_ads_data.info())

# Basic Statistics
print("\nStatistik Dasar:")
print(marketing_ads_data.describe())

# Missing Values Check
print("\nMissing Values:")
print(marketing_ads_data.isnull().sum())

# DUplicate Check
```

```
print("\nDuplikat:")
print(marketing_ads_data.duplicated().sum())

# Unique Values Check for Important Columns
print("\n--- Unique Values Check for Important Columns:---")
print("Variant:", marketing_ads_data['variant'].unique())
print("Campaign Name:", marketing_ads_data['campaign_name'].unique())
print("Audience Type:", marketing_ads_data['audience_type'].unique())
print("Product Type:", marketing_ads_data['product_type'].unique())
print("Product Name:", marketing_ads_data['product_name'].unique())

# Data Sample
print("Data Sample:", marketing_ads_data.sample(10))


print("\n===== SALES TRANSACTION DATA =====")

# Data Structure
print("\nStruktur Data:")
print(sales_transaction_data.info())

# Basic Statistics
print("\nStatistik Dasar:")
print(sales_transaction_data.describe())

# Missing Values Check
print("\nMissing Values:")
print(sales_transaction_data.isnull().sum())
```

```
# DUplicate Check
print("\nDuplikat:")
print(sales_transaction_data.duplicated().sum())

# Unique Values Check for Important Columns
print("\n--- Unique Values Check for Important Columns:---")
print("Variant:", sales_transaction_data['variant'].unique())
print("Product Type:", sales_transaction_data['product_type'].unique())
print("Product Name:", sales_transaction_data['product_name'].unique())
print("Order Status:", sales_transaction_data['order_status'].unique())

# Data Sample
print("Data Sample:", sales_transaction_data.sample(10))
```

```
===== MARKETING ADS DATA =====

Struktur Data:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1030 entries, 0 to 1029
Data columns (total 12 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   ad_id          1030 non-null   object
 1   campaign_name  1030 non-null   object
 2   audience_type  1030 non-null   object
 3   product_type   1030 non-null   object
 4   product_name   1030 non-null   object
 5   variant        1030 non-null   object
 6   date           1030 non-null   object
 7   impressions    1030 non-null   int64
 8   clicks         1030 non-null   int64
 9   spend_idr      970 non-null    float64
 10  sessions       1030 non-null   int64
 11  add_to_cart    1030 non-null   int64
dtypes: float64(1), int64(4), object(7)
memory usage: 96.7+ KB
None

Statistik Dasar:
        impressions       clicks     spend_idr     sessions  add_to_cart
count   1030.000000  1030.000000  9.700000e+02  1030.000000  1030.000000
mean   12302.346602   228.747573  3.095047e+05   194.000971    23.620388
std     3796.802668   122.200596  1.934729e+05   104.467101    16.449072
min     5078.000000    47.000000  5.048000e+03    38.000000     3.000000
25%     9206.250000   135.000000  1.975175e+05   115.000000    12.000000
50%    12013.000000   200.500000  3.002635e+05   171.000000    19.000000
75%    15052.250000   296.500000  4.045185e+05   250.750000    31.000000
max    19940.000000   624.000000  2.839487e+06   536.000000    99.000000

Missing Values:
ad_id             0
campaign_name     0
audience_type     0
product_type      0
product_name      0
variant           0
date              0
impressions       0
clicks            0
spend_idr        60
sessions          0
add_to_cart       0
dtype: int64

Duplikat:
30

--- Unique Values Check for Important Columns:---
Variant: ['Video_Review' 'Static_Catalog']
Campaign Name: ['Awareness Campaign' 'Retargeting Campaign' 'Flash Sale Campaign']
Audience Type: ['cold' 'warm']
Product Type: ['Serum' 'Sunscreen' 'Cleanser' 'Moisturizer']
Product Name: ['DermaGlow Brightening Serum' 'DermaGlow UV Shield Sunscreen'
 'DermaGlow Gentle Cleanser' 'DermaGlow Hydrating Moisturizer']
Data Sample:        ad_id         campaign_name audience_type product_type  \
414  AD_0007  Retargeting Campaign          warm        Serum
597  AD_0150   Flash Sale Campaign          cold        Serum
198  AD_0990   Flash Sale Campaign          cold        Serum
988  AD_0567    Awareness Campaign          cold    Sunscreen
559  AD_0435    Awareness Campaign          cold    Sunscreen
579  AD_0705   Flash Sale Campaign          cold    Sunscreen
737  AD_0148    Awareness Campaign          cold    Sunscreen
37   AD_0970   Flash Sale Campaign          cold    Sunscreen
852  AD_0564   Flash Sale Campaign          cold    Sunscreen
122  AD_0353  Retargeting Campaign          warm     Cleanser

                     product_name         variant          date  impressions  \
414    DermaGlow Brightening Serum  Static_Catalog    2023-11-06        13873
597    DermaGlow Brightening Serum  Static_Catalog    08/09/2023        16105
198    DermaGlow Brightening Serum    Video_Review    2023-05-16        14192
988  DermaGlow UV Shield Sunscreen    Video_Review    2023-03-26        10982
559  DermaGlow UV Shield Sunscreen    Video_Review    2023-04-02        13411
579  DermaGlow UV Shield Sunscreen  Static_Catalog    2023-06-15        14630
737  DermaGlow UV Shield Sunscreen    Video_Review  Nov 25, 2023         6380
37   DermaGlow UV Shield Sunscreen    Video_Review    2023-10-12        11284
852  DermaGlow UV Shield Sunscreen    Video_Review    2023-09-03        14974
122      DermaGlow Gentle Cleanser    Video_Review    2023-06-30         9222

     clicks  spend_idr  sessions  add_to_cart
414     161   137065.0       143           19
597     144   488411.0       116           21
```

```
198     404   142678.0       350           26
988     281        NaN       230           33
559     313   208944.0       258           36
579     163   442016.0       142            8
737     120   183560.0       100            9
37      351   372217.0       283           19
852     312   255918.0       268           14
122     273   486721.0       223           25

===== SALES TRANSACTION DATA =====

Struktur Data:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2500 entries, 0 to 2499
Data columns (total 8 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   transaction_id    2500 non-null   object
 1   ad_id             2408 non-null   object
 2   transaction_date  2500 non-null   object
 3   product_type      2500 non-null   object
 4   product_name      2500 non-null   object
 5   variant           2500 non-null   object
 6   revenue_idr       2500 non-null   float64
 7   order_status      2500 non-null   object
dtypes: float64(1), object(7)
memory usage: 156.4+ KB
None

Statistik Dasar:
        revenue_idr
count  2.500000e+03
mean   9.273054e+05
std    1.650125e+07
min   -4.450000e+05
25%    8.675000e+04
50%    2.175000e+05
75%    3.580000e+05
max    4.692353e+08

Missing Values:
transaction_id       0
ad_id               92
transaction_date     0
product_type         0
product_name         0
variant              0
revenue_idr          0
order_status         0
dtype: int64

Duplikat:
0

--- Unique Values Check for Important Columns:---
Variant: ['Static_Catalog' 'Video_Review']
Product Type: ['Cleanser' 'Moisturizer' 'Serum' 'Sunscreen']
Product Name: ['DermaGlow Gentle Cleanser' 'DermaGlow Hydrating Moisturizer'
 'DermaGlow Brightening Serum' 'DermaGlow UV Shield Sunscreen']
Order Status: ['Success' 'Cancelled' 'Refunding']
Data Sample:      transaction_id    ad_id transaction_date product_type  \
51       TXN_001756  AD_0440       2023-09-28        Serum
160      TXN_001365  AD_0429       2023-05-19    Sunscreen
2063     TXN_001904  AD_0084       2023-12-29    Sunscreen
2363     TXN_001664  AD_0602       2023-06-23  Moisturizer
937      TXN_002226  AD_0359       2023-12-26  Moisturizer
933      TXN_001906  AD_0084       2023-12-29    Sunscreen
720      TXN_000314  AD_0469       2023-02-09        Serum
2240     TXN_001046      NaN       2023-03-17  Moisturizer
689      TXN_002366  AD_0361       2023-02-26    Sunscreen
1772     TXN_002101  AD_0812       2023-08-12    Sunscreen

                        product_name         variant  revenue_idr  \
51        DermaGlow Brightening Serum  Static_Catalog     230000.0
160     DermaGlow UV Shield Sunscreen    Video_Review     306000.0
2063    DermaGlow UV Shield Sunscreen  Static_Catalog    -122000.0
2363  DermaGlow Hydrating Moisturizer    Video_Review     304000.0
937   DermaGlow Hydrating Moisturizer    Video_Review     229000.0
933     DermaGlow UV Shield Sunscreen  Static_Catalog     350000.0
720       DermaGlow Brightening Serum  Static_Catalog          0.0
2240  DermaGlow Hydrating Moisturizer  Static_Catalog          0.0
689     DermaGlow UV Shield Sunscreen  Static_Catalog          0.0
1772    DermaGlow UV Shield Sunscreen  Static_Catalog     343000.0

     order_status
51        Success
160       Success
```

```
2063    Refunding
2363      Success
937       Success
933       Success
720     Cancelled
2240    Cancelled
689     Cancelled
1772      Success
```

FINDINGS:

- **Marketing Ads 3 masalah**
  - Kolom Spend IDR = 60 nulls (5.8% dari data)
  - Kolom Date format masih object, belum datetime
  - 30 duplikat
- **Sales Transaction 3 masalah**
  - Kolom Date format masih object, belum datetime
  - Kolom Revenue IDR ada nilai negatif (min: -445.000) dan nilai (max: 469 juta) kemungkinan outlier.
  - Kolom ad_id 92 null transaksi tanpa tracking
  - Kolom Revenue IDR ada nilai 0.0, kemungkinan status order cancelled, perlu konfirmasi.

In [3]:
```
# CELL 2
# DATA CLEANING & TRANSFORMATION

# MARKETING ADS DATA

# 1. Drop duplicates
marketing_ads_data = marketing_ads_data.drop_duplicates()
print("Marketing Ads setelah drop duplicates:", marketing_ads_data.shape)

# 2. Standarisasi kolom Date
marketing_ads_data['date'] = pd.to_datetime(
    marketing_ads_data['date'], format='mixed', dayfirst=True
)
print("Tipe data kolom date:", marketing_ads_data['date'].dtype)

# 3. Investigasi pola null sebelum treatment
print("\n === Null spend_idr per campaign_name: ===")
print(marketing_ads_data[marketing_ads_data['spend_idr'].isnull()]['campaign_name'].value_counts())

print("\n === Null spend_idr per audience_type: ===")
print(marketing_ads_data[marketing_ads_data['spend_idr'].isnull()]['audience_type'].value_counts())

# SALES TRANSACTIONS DATA

# 1. Standarisasi kolom Date
sales_transaction_data['date'] = pd.to_datetime(sales_transaction_data['transaction_date'], format='mixed', dayfirst=True)
print("Tipe data kolom date:", sales_transaction_data['date'].dtype)

# 2. investigasi kolom revenue_idr negatif
print("\n === revenue_idr negatif per order_status ===")
print(sales_transaction_data[sales_transaction_data['revenue_idr'] < 0]['order_status'].value_counts())
print("Jumlah baris revenue negatif:", sales_transaction_data[sales_transaction_data['revenue_idr'] < 0].shape[0])

# 3. Investigasi revenue outlier ekstrim (> 10 juta)
print("\n === Revenue ekstrim (>10 juta) ===")
print(sales_transaction_data[sales_transaction_data['revenue_idr'] > 10_000_000][['transaction_id', 'revenue_idr', 'order_stat

# Drop kolom date karena sudah ada kolom transaction_date yang lebih spesifik
sales_transaction_data = sales_transaction_data.drop(columns=['date'])
print("\nSales Transactions columns setelah drop date:")
print(sales_transaction_data.columns.tolist())
```

```
Marketing Ads setelah drop duplicates: (1000, 12)
Tipe data kolom date: datetime64[ns]

 === Null spend_idr per campaign_name: ===
campaign_name
Flash Sale Campaign     26
Awareness Campaign      17
Retargeting Campaign    13
Name: count, dtype: int64

 === Null spend_idr per audience_type: ===
audience_type
cold    43
warm    13
Name: count, dtype: int64
Tipe data kolom date: datetime64[ns]

 === revenue_idr negatif per order_status ===
order_status
Refunding    209
Name: count, dtype: int64
Jumlah baris revenue negatif: 209

 === Revenue ekstrim (>10 juta) ===
     transaction_id  revenue_idr order_status
708       TXN_001591  266506890.0      Success
945       TXN_001160  348177883.0      Success
1393      TXN_001980  299801663.0      Success
2227      TXN_000547  424370733.0      Success
2410      TXN_000419  469235332.0      Success

Sales Transactions columns setelah drop date:
['transaction_id', 'ad_id', 'transaction_date', 'product_type', 'product_name', 'variant', 'revenue_idr', 'order_status']
```

CELL 2 - DATA CLEANING OUTPUT

MARKETING ADS - kolom spend_idr 56 nulls:

Null tersebar di semua campaign (Flash Sale: 26, Awareness Campaign: 17, Retargeting Campaign: 13), mayoritas pada Cold audience (43) kemungkinan missing at random. Decision: fill null dengan median per campaign_name (lebih robust dari mean karna ada outlier di spend).

SALES TRANSACTION DATA - 3 Decisions

- revenue_idr negatif semua dari Refunding (209) valid secara bisnis, tidak saya drop tapi akan saya konversi ke absolut.
- 5 baris revenue ekstrim (266 juta - 469 juta) dengan status success, asumsi merupakan data anomali dan bukan transaksi normal produk skincare, decision: drop.
- kolom ad_id null 92 baris tidak di drop karena transaksi tetap valid untuk analisis revenue.

In [4]:
```
# CELL 3
# DATA TREATMENT

# MARKETING ADS DATA

# Fill null spend_idr dengan median per campaign_name
marketing_ads_data['spend_idr'] = marketing_ads_data.groupby('campaign_name')['spend_idr'].transform(lambda x: x.fillna(x.medi
print("Null spend_idr setelah fill:", marketing_ads_data['spend_idr'].isnull().sum())

# SALES TRANSACTIONS DATA

# 1. Drop revenue ekstrim
before = sales_transaction_data.shape[0]
sales_transaction_data = sales_transaction_data[sales_transaction_data['revenue_idr'] <= 10_000_000]
print(f"Baris di drop karna revenue ekstrim: {before - sales_transaction_data.shape[0]}")

# 2. Revenue negatif Refunding di konversi ke absolut
sales_transaction_data['revenue_idr'] = sales_transaction_data['revenue_idr'].abs()
print("Min revenue_idr setelah konversi:", sales_transaction_data['revenue_idr'].min())

# FINAL CHECK

print("\n=== Final Shape ===")
print("Marketing Ads:", marketing_ads_data.shape)
print("Sales Transactions:", sales_transaction_data.shape)

print("\n=== Final Null Check ===")
print(marketing_ads_data.isnull().sum())
print(sales_transaction_data.isnull().sum())
```

```
Null spend_idr setelah fill: 0
Baris di drop karna revenue ekstrim: 5
Min revenue_idr setelah konversi: 0.0

=== Final Shape ===
Marketing Ads: (1000, 12)
Sales Transactions: (2495, 8)

=== Final Null Check ===
ad_id            0
campaign_name    0
audience_type    0
product_type     0
product_name     0
variant          0
date             0
impressions      0
clicks           0
spend_idr        0
sessions         0
add_to_cart      0
dtype: int64
transaction_id       0
ad_id               92
transaction_date     0
product_type         0
product_name         0
variant              0
revenue_idr          0
order_status         0
dtype: int64
```

CELL 3

SUMMARY

1. kolom spend_idr null diisi dengan nilai median per campaign tidak ada lagi kolom null tersisa.
2. revenue ekstrim sudah tidak ada.
3. revenue negatif di konversi ke absolut.
4. Shape Final: Marketing Ads: (1000, 12), Sales Transactions: (2495, 8).

In [5]:
```python
# CELL 4
# MERGE DATA

# 1. Agregasi revenue per ad_id dari sales transaction. Hanya ambil transaksi Success saja untuk revenue yang valid
revenue_per_ad = sales_transaction_data[sales_transaction_data['order_status'] == 'Success'].groupby('ad_id').agg(
    total_revenue=('revenue_idr', 'sum'),
    total_orders=('transaction_id', 'count')
).reset_index()

print("Shape revenue_per_ad:", revenue_per_ad.shape)
print(revenue_per_ad.head())

# 2. Merge dengan marketing ads data
merged_data = marketing_ads_data.merge(revenue_per_ad, on='ad_id', how='left')

print("\nShape merged_data:", merged_data.shape)
print(merged_data.isnull().sum())

# 3. Fill null total_revenue dan total_orders dengan 0
merged_data[['total_revenue', 'total_orders']] = merged_data[['total_revenue', 'total_orders']].fillna(0)

print("Null setelah fill:")
print(merged_data[['total_revenue', 'total_orders']].isnull().sum())
print("Shape:", merged_data.shape)
```

```
Shape revenue_per_ad: (587, 3)
     ad_id  total_revenue  total_orders
0  AD_0001      1450000.0             5
1  AD_0002       723000.0             2
2  AD_0004       806000.0             3
3  AD_0006       636000.0             2
4  AD_0007      1045000.0             3

Shape merged_data: (1000, 14)
ad_id              0
campaign_name      0
audience_type      0
product_type       0
product_name       0
variant            0
date               0
impressions        0
clicks             0
spend_idr          0
sessions           0
add_to_cart        0
total_revenue    413
total_orders     413
dtype: int64
Null setelah fill:
total_revenue    0
total_orders     0
dtype: int64
Shape: (1000, 14)
```

CELL 4 (MERGE DATA) - INSIGHT & KEY FINDINGS

Agregasi dilakukan sebelum Merge Data karena pada kasus ini, kolom 'ad_id' di data Sales Transactions bisa banyak transaksi per Ad. Revenue perlu agregasi per Ad di Sales Transaction sebelum merge agar tidak terjadi row explosion.

Yang Saya Lakukan:

Melakukan agregasi sebelum merge data. Join nya adalah Marketing Ads Data LEFT JOIN Sales Transaction Data on ad_id agar semua data Ad tetap terhitung meski tanpa transaksi.

In [6]:
```python
# CELL 5
# KALKULASI METRIK

merged_data['CTR']  = merged_data['clicks'] / merged_data['impressions']
merged_data['CPC']  = merged_data['spend_idr'] / merged_data['clicks']
merged_data['ROAS'] = merged_data['total_revenue'] / merged_data['spend_idr']
merged_data['CVR']  = merged_data['total_orders'] / merged_data['sessions']

print("=== Sample Metrics ===")
print(merged_data[['ad_id', 'CTR', 'CPC', 'ROAS', 'CVR']].head(10))

print("\n=== Statistic Metric per Variant ===")
print(merged_data.groupby('variant')[['CTR', 'CPC', 'ROAS', 'CVR']].mean().round(4))
```

```
=== Sample Metrics ===
     ad_id       CTR          CPC       ROAS       CVR
0  AD_0032  0.031390  1813.818182   0.000000  0.000000
1  AD_0110  0.031304   199.391144   0.000000  0.000000
2  AD_0137  0.010993   752.416667  17.240743  0.054795
3  AD_0089  0.014811  1535.333333   7.144086  0.054688
4  AD_0919  0.021533  1147.164179   1.766198  0.004608
5  AD_0169  0.032263   608.647856   5.600246  0.011268
6  AD_0871  0.025243  1058.367713   3.779405  0.011236
7  AD_0319  0.021347   925.897106   0.000000  0.000000
8  AD_0262  0.014821  1865.315789   5.111110  0.042017
9  AD_0536  0.014141  2570.496063   0.000000  0.000000

=== Statistic Metric per Variant ===
                  CTR        CPC    ROAS     CVR
variant
Static_Catalog  0.0119  2538.6264  3.4047  0.0191
Video_Review    0.0250  1117.3851  1.7696  0.0051
                  CTR        CPC    ROAS     CVR
variant
Static_Catalog  0.0119  2538.6264  3.4047  0.0191
Video_Review    0.0250  1117.3851  1.7696  0.0051
```

CELL 5 (KALKULASI METRIK) - INSIGHT & KEY FINDINGS

Yang Saya Lakukan:

Melakukan kalkulasi untuk metrik yang dibutuhkan seperti CPC, CVR, ROAS dan CTR.

Key Findings:

- **Finding 1:** CTR: Video Review unggul (2% vs 1%), orang lebih banyak klik video daripada catalog
- **Finding 2:** CPC: Video Review lebih efisien secara cost per click (Rp. 1,117 vs Rp. 2,538).
- **Finding 3:** ROAS: Static Catalog unggul (3% vs 2%), indikasi audience pada Catalog memiliki purchase intent yang tinggi.
- **Finding 4:** CVR: Kedua variant hampir sama persis (Static catalog 0.8459 vs Video Review 0.8481)

Perbedaan ini akan diuji signifikansinya pada CELL selanjutnya

In [7]:
```
# CELL 6 - EXPLORATORY DATA ANALYSIS (EDA)
# STATISTICAL SIGINIFCANCE TESTING (SHAPIRO WILK TEST & MANN WHITNEY U TEST)

from scipy import stats

# 1. Data per Variant dipisahkan dahulu
video   = merged_data[merged_data['variant'] == 'Video_Review']
static  = merged_data[merged_data['variant'] == 'Static_Catalog']

# 2. Cek normalitas menggunakan Shapiro Wilk Test
print("=== Shapiro Wilk Test for Normality ===")
for metric in ['CTR', 'CVR', 'CPC', 'ROAS']:
    _, p_video = stats.shapiro(video[metric])
    _, p_static = stats.shapiro(static[metric])
    print(f"{metric} — Video p: {p_video:.4f} | Static p: {p_static:.4f}")

# 3. Cek signifikansi perbedaan antara kedua variant menggunakan Mann Whitney U Test (karena data tidak normal)
print("\n=== Mann-Whitney U Test ===")
for metric in ['CTR', 'CVR', 'CPC', 'ROAS']:
    stat, p_value = stats.mannwhitneyu(
        video[metric], static[metric], alternative='two-sided'
    )
    significance = "SIGNIFICANT" if p_value < 0.05 else "not significant"
    print(f"{metric} — p-value: {p_value:.4f} | {significance}")
```

```
=== Shapiro Wilk Test for Normality ===
CTR — Video p: 0.0000 | Static p: 0.0000
CVR — Video p: 0.0000 | Static p: 0.0000
CPC — Video p: 0.0000 | Static p: 0.0000
ROAS — Video p: 0.0000 | Static p: 0.0000

=== Mann-Whitney U Test ===
CTR — p-value: 0.0000 | SIGNIFICANT
CVR — p-value: 0.0000 | SIGNIFICANT
CPC — p-value: 0.0000 | SIGNIFICANT
ROAS — p-value: 0.0001 | SIGNIFICANT
```

CELL 6 MARKDOWN

Findings:

- Shapiro-Wilk menunjukkan semua metrik tidak berdistribusi normal, Mann-Whitney U dipilih sebagai non parametric alternatif
- Semua metrik (CTR, CVR, CPC, ROAS) menunjukkan perbedaan yang signifikan secara statistik antara Video_Review dan Static_Catalog ($p < 0.05$).

In [8]:
```
# CELL 7 - EXPLORATORY DATA ANALYSIS (EDA)
# SEGMENT ANALYSIS

# 1. Per Audience Type
print("=== Segment Analysis: Audience Type ===")
print(merged_data.groupby(['audience_type', 'variant'])[['CTR', 'CVR', 'CPC', 'ROAS']].mean().round(4))

# 2. Per Product Type
print("\n === Segment Analysis: Product Type ===")
print(merged_data.groupby(['product_type', 'variant'])[['CTR', 'CVR', 'CPC', 'ROAS']].mean().round(4))
```

```
=== Segment Analysis: Audience Type ===
                                CTR     CVR        CPC    ROAS
audience_type variant
cold          Static_Catalog  0.0120  0.0180  2568.7309  2.9863
              Video_Review    0.0250  0.0050  1145.0309  1.7150
warm          Static_Catalog  0.0117  0.0213  2479.4798  4.2268
              Video_Review    0.0252  0.0053  1058.7901  1.8855

 === Segment Analysis: Product Type ===
                                CTR     CVR        CPC    ROAS
product_type variant
Cleanser     Static_Catalog  0.0119  0.0196  2574.7858  2.2872
             Video_Review    0.0247  0.0065  1011.8212  2.1635
Moisturizer  Static_Catalog  0.0116  0.0181  2527.2755  3.0546
             Video_Review    0.0248  0.0042  1215.6686  1.3945
Serum        Static_Catalog  0.0120  0.0187  2597.6584  3.6989
             Video_Review    0.0251  0.0049  1097.4098  2.0389
Sunscreen    Static_Catalog  0.0121  0.0201  2463.4566  4.3527
             Video_Review    0.0256  0.0051  1111.5418  1.6006
```

CELL 7 - MARKDOWN

INSIGHT AND FINDINGS

Findings:

- **Finding 1:** Video_Review selalu unggul di CTR dan CPC pada semua segment
- **Finding 2:** Static_Catalog selalu unggul di CVR dan ROAS
- **Finding 3:** Warm audience static catalog tertinggi (4.23), paling tinggi di semua segment
- **Finding 4:** Sunscreen static catalog ROAS tertinggi (4.35), produk return paling tinggi
- **Finding 5:** Moisturizer video review ROAS buruk di Video Review (1.40).

Insight:

Video Review efisien pada top funnel sedangkan Static Catalog efektif pada bottom funnel, indikasi Video berfungsi untuk mendapatkan attention dan kemudian conversion melalui Static Catalog. Sunscreen menjadi prioritas utama campaign karena menghasilkan return tertinggi dalam segment product type.

CELL 8 - BUSINESS QUESTION & RECCOMMENDATION

Business Question 1: Variant mana yang perform lebih baik overall?

Tidak ada 1 variant yang unggul di semua metrik, kedua variant unggul di area yang berbeda - Video Review: Unggul di Top Funnel: CTR (0.0250 vs 0.0119), CPC (Rp. 1,117 vs Rp. 2,538). - Static Catalog: Unggul di Bottom Funnel: CVR (0.8459 vs 0.8481), ROAS (3.49 vs 1.77).

Business Question 2: Apakah perbedaan performa tersebut signifikan secara statistik?

Berdasarkan hasil testing dengan MANN WHITNEY U TEST menunjukkan perbedaan secara signifikan ($p < 0.05$) di semua metrik. **Metrik:** - CTR : < 0.0001 Signifikan - CVR : < 0.0001 Signifikan - CPC : < 0.0001 Signifikan - ROAS : < 0.0001 SIgnifikan Perbedaan performa bukan kebetulan, valid menjadi dasar keputusan bisnis.

Business Question 3: Bagaimana performa masing - masing variant di berbagai segment?

1. **Audience Type**
    - Warm Audience konsisten menghasilkan ROAS lebih tinggi di kedua variant, Static Catalog Warm Audience menacapai ROAS 4.23 tertinggi dari semua segment audience
    - Cold Audience lebih responsif terhadap Video Review dari sisi CTR namun ROAS tetap lebih rendah dari Static Catalog.
2. **Product Type**
    - Sunscreen adalah produk dengan ROAS tertinggi di variant Static Catalog (4.35)
    - Moisturizer adalah kombinasi terlemah pada variant Video Review, ROAS (1.35)
    - Static Catalog konsisten unggul ROAS di semua product type.

Business Question 4: Apakah ada trade off antara CTR dan Conversion Rate?

Trade Off konsisten di semua segment. - Video Review menarik lebih banyak klik (CTR tinggi) namun menghasilkan revenue per spend yang lebih rendah (ROAS rendah) - Static Catalog menghasilkan lebih sedikit klik namun setiap klik yang dihasilkan berkualitas, dibuktikan dengan ROAS yang lebih tinggi - Ini mengindikasikan bahwa audience yang tertarik lewat variant Static Catalog memiliki purchase intent lebih tinggi

Business Question 5: Dari segi bisnis, mana lebih layak di scale?

Rekomendasi hybrid strategy.

Tidak disarankan memilih salah satu variant secara ekslusif.

1. Scale Static Catalog untuk:
    - Warm Audience, ROAS 4.23 purchase intent tinggi
    - Produk Sunscreen dan Serum, ROAS tertinggi di produk ini
    - Campaign yang berfokus pada revenue dan profit
2. Scale Video Review untuk:
    - Cold Audience, efektif untuk menarik audience baru dengan CPC lebih efisien
    - Top of Funnel campaign untuk reach dan engagement bukan immediate conversion.

Next Step yang disarankan

1. ALokasikan budget lebih besar pada Static Catalog untuk warm audience
2. Optimalkan landing page untuk Video Review, CVR nya rendah padahaln traffic tinggi, kemungkinan ada friction di post click experience
3. Lakukan AB test lanjutan untuk moisturizer, kombinasi Video Review + Moisturizer paling underperform dan perlu investigasi lebih lanjut.

In [9]:
```
# CELL 9 - EXPORT DATASET

merged_data.to_csv("dermaglow_merged_data_cleaned.csv", index=False)
print("Data Exported:", merged_data.shape)
```

```
Data Exported: (1000, 18)
 (1000, 18)
```